JEYA VIKNAN

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SUMMARY

Full-stack engineer and **co-founder of HUDO** who takes software from client requirements to production release. Ships web products in **React, Next.js and TypeScript** on **Node.js, Python (FastAPI)** and Spring Boot services with PostgreSQL and AWS, and builds **LLM and RAG** features into logistics, finance and operations workflows. Built two campus tools in **active use** by VIT students and faculty. **CodeChef 6-star (India Rank 199)**; **AWS Certified Solutions Architect – Associate**.

EDUCATION

Vellore Institute of Technology, Chennai 2023 – May 2027
B.Tech, Computer Science and Engineering | CGPA: 8.5/10

Indian Institute of Technology Madras 2023 – 2027
BS, Data Science and Applications (online, concurrent) | CGPA: 8.0/10

Coursework: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

Co-Founder and Product Engineer | HUDO, Software Services Startup 2025 – Present

- Co-founded a software-services startup that builds custom software to each client's needs; delivered **3 client products**, each closed with **positive client feedback**, and currently lead **2 active engagements**.
- Own every engagement end to end: translate client pain points into scoped features with clear acceptance criteria, then drive delivery in **Jira** through backlog prioritisation, sprint planning and blocker tracking.

AI Engineering Intern | Integra Global Solutions, Coimbatore May 2026 – Jul 2026

- Cut document retrieval time by **40%** across **5,000+ documents** by building a production **RAG knowledge assistant** for accounting and BPO operations teams, as one of a three-engineer team.
- Built its ETL pipeline (ingestion, cleaning, chunking, embedding) and **FastAPI** microservices with token auth, role-based access and OpenAPI docs, behind a **React** interface.
- Reduced release cycle time by **30%** with a **GitHub Actions** CI/CD pipeline that runs PyTest unit and integration suites on every commit and ships Dockerised services.

AI Research Intern | National Institute of Technology, Tiruchirappalli May 2025 – Jul 2025

- Achieved **86.8% accuracy** on a **2,000+ sample** sarcasm-detection dataset with an end-to-end Python and scikit-learn pipeline covering cleaning, feature engineering, training and automated evaluation.

PROJECTS

Meridian — Last-Mile Delivery Platform Next.js, React, TypeScript, PostgreSQL, Prisma | Live | GitHub

- Shipped a live **three-role logistics platform** (customer, delivery agent, operations admin) with five-step booking and parcel tracking, built as a Next.js modular monolith over a framework-free, unit-tested domain layer.
- Built an explainable **rate engine** (volumetric weight, admin-configured slab rate cards, COD surcharge) that shows a line-by-line price breakdown, with money held in integer paise so no float ever touches a charge.
- Made dispatch **race-free**: agents are ranked on zone, distance and live load, and capacity is claimed with one guarded `UPDATE`, so **6 concurrent claims** on a capacity of 2 yield exactly 2 winners.
- Turned failed deliveries into **customer self-rescheduling** with automatic reassignment, recorded on a trigger-enforced **append-only** timeline; covered by **252 automated tests**, 104 of them integration tests on real PostgreSQL.

Switchyard — Multi-Tenant LLM Gateway Python, Prometheus, Grafana, Docker | GitHub

- Built a self-hostable, **OpenAI-compatible** gateway that shares scarce LLM inference capacity fairly across tenants through **token-weighted fair scheduling**, reserved floors, ceilings and hard per-tenant token budgets.
- Kept it stable under overload and provider failure with bounded-queue admission control, deadline-based load shedding, phased timeouts and **circuit-breaker failover** that never splices partial answers.
- Made every scheduling decision observable through **Prometheus** metrics and an 11-panel **Grafana** dashboard, and proved behaviour with runnable noisy-neighbour and provider-outage scenarios.

Penny — Double-Entry Banking Ledger Java 17, Spring Boot 3, PostgreSQL, React | GitHub

- Held **zero balance drift** under a **50-thread** concurrent-transfer test using ordered `SELECT FOR UPDATE` locks and idempotency keys, with balances derived from a trigger-enforced, append-only journal.
- Secured with Spring Security and JWT across **4 role tiers**; every merge gated at **85% test coverage** (JUnit 5, Testcontainers, JaCoCo) in a 3-job GitHub Actions pipeline.

TOOLS IN ACTIVE USE

Sorry No Proxy — Proxy-Resistant Attendance Node.js, Python, React | GitHub

- Adopted by **many VIT faculty members** as their go-to attendance system, sustaining **500+ daily verifications**; submitted a proposal to make it **VIT's default attendance platform**.
- Closed three proxy paths (screenshots, leaked page secrets, clock tampering) with server-signed QR codes rotating every **400 ms**, single-use device-bound tokens and multi-frame face matching; **104 automated tests**.

Nankiv — Placement Shortlist Tracker

- Built a tool that lets final-year students monitor and manage their company shortlists across campus drives in one place; **widely used by VIT final-year students** through placement season.

RESEARCH AND PATENTS

- **Soft-Membership Community Indexing for GraphRAG** (final-year research): decomposed GraphRAG's index-time information loss into three measurable components to motivate a new indexing method.
- **HC-WF, Encrypted Traffic Classification** (sole author, manuscript in preparation): two-stage PyTorch Transformer reaching **92.4%** closed-world accuracy and **84.6%** open-world F1 over published baselines.
- **F.O.R.G.E** (co-inventor, patent filing in progress): fog-orchestrated fire-evacuation system on Raspberry Pi, Arduino and IoT sensors that routes people using live fire-location and occupancy data.

LEADERSHIP AND ACHIEVEMENTS

- **Technical Head, VITeach** (Dec 2024 – Present): lead a **10+ developer** team shipping platforms used by **1,000+ students**; introduced Git branching conventions, mandatory PR review and CI/CD gates.
- **CodeChef 6-star** (rating 2200+, Division 1) with **India Rank 199** and **Global Rank 333**; **150+** problems solved on LeetCode.
- Shortlisted to represent VIT Chennai at the **Smart India Hackathon**.

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate**, Amazon Web Services (Aug 2026 – Aug 2029) | Credential
- **Atlas Vector Search and RAG Applications**, MongoDB University (Mar 2026) | Credential
- **Prompt Engineering Track**, Anthropic Academy (Sep 2026) | Credential

SKILLS

Languages: TypeScript, JavaScript, Python, Java, SQL, C++

Frontend: React, Next.js, Tailwind CSS, TanStack Query, HTML, CSS

Backend and APIs: Node.js, FastAPI, Spring Boot, Spring Security, REST APIs, Microservices, Prisma, Kafka

AI and Data: LLMs, RAG, Prompt Engineering, PyTorch, HuggingFace, FAISS, scikit-learn, pandas, NumPy, ETL

Databases: PostgreSQL, MySQL, MongoDB, Redis

Cloud and DevOps: AWS (EC2, ECS, ECR, S3, IAM), Docker, Kubernetes, GitHub Actions, CI/CD, Prometheus, Grafana, Linux, Git

Testing and Delivery: Vitest, JUnit 5, Testcontainers, PyTest; Jira, Agile/Scrum, requirements gathering

Core CS: Data Structures and Algorithms, OOP, System Design (HLD/LLD), DBMS, Operating Systems, Networks